Conversion of NFA to DFA:

Note: Both NFA and DFA are equally powerful.

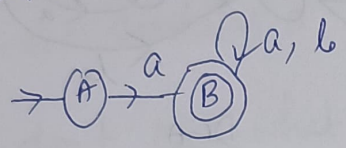
- Therefore, every NFA can be converted to a DFA and every DFA can be converted to an NFA. Hence every DFA is a NFA by its definition, so, we need to only know how to convert a NFA to DFA.

If an NFA has n states, its equivalent DFA can have at most 2^n states

Ex: ① Take a Language, L = set of all strings starts with an "a" over

$\Sigma = \{a, b\}$

then NFA looks like as follows (It should starts with "a" and we don't care what happens after that.



- Now, we convert this NFA to DFA, however there are various methods to convert an NFA to DFA. Here we are going to follow "Subset Construction" method.

Step1: State transition Table of this NFA

	a	b
→ A	{B}	Φ
* B	{B}	{B}

	a	b
→ A	[B]	[D]
* B	[B]	[B]
D	[D]	[D]

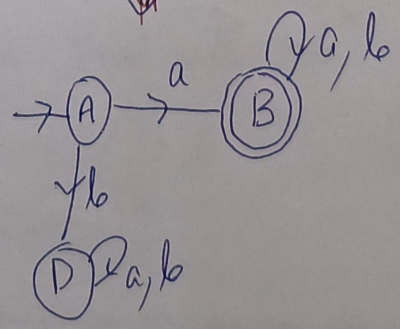
(dead state in DFA)

(because on seeing "a" or "b", we should reside in state "D".

or

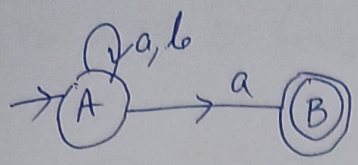
	a	b
→ A	[B]	Φ
* B	[B]	[B]
Φ	Φ	Φ

draw the dfa



Conversion of NFA to DFA

Ex: (2). L = all strings ends with "a" over {a, b}



NFA (here initially, I don't consider whatever is coming "a" or "b" but, it must end with "a", So)

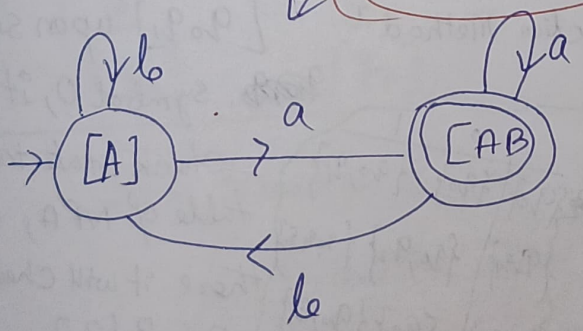
	a	b
→ A	{A, B}	{A}
*B	{ }	{ }

⇒

	a	b
→ [A]	[A, B]	[A]
* [A, B]	[A, B]	[A]

Note: we need not to write [B] at second row, Instead, we write [AB] state, because only two states ([AB] and [A]) are reachable from state [A] at the first row.

draw the dfa



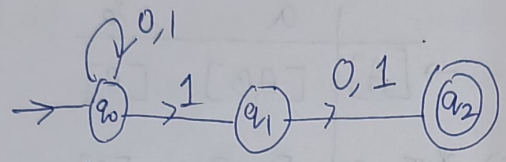
Conversion of NFA to DFA

Ex: (3) An NFA accepting the set of all strings whose second last symbol is 1 over alphabet $\Sigma = \{0, 1\}$

or

An NFA accepting the set of all string whose 2nd Symbol from R.H.S is 1 over $\Sigma = \{0, 1\}$

Ans



State transition diagram of NFA

⇓ get state transition table

	0	1
→ q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
* q_2	—	—

State transition table of NFA

or

	0	1
→ q_0	q_0	q_0, q_1
q_1	q_2	q_2
* q_2	Φ	Φ

(Instead of "—", you can also write $\{\}$ or Φ)

⇓ Use "subset Construction Method"

	0	1
→ $[q_0]$	$[q_0]$	$[q_0, q_1]$
$[q_0, q_1]$	$[q_0, q_2]$	$[q_0, q_1, q_2]$
* $[q_0, q_2]$	$[q_0]$	$[q_0, q_1]$
* $[q_0, q_1, q_2]$	$[q_0, q_2]$	$[q_0, q_1, q_2]$

or

	0	1
→ $\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$
* $\{q_0, q_2\}$	$\{q_0\}$	$\{q_0, q_1\}$
* $\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$

$[q_0 q_1]$ upon seeing symbol 0, it will check state transition table of NFA, and there it will check

$q_0 \xrightarrow{0} \{q_0\}$

$q_1 \xrightarrow{0} \{q_2\}$

and take $\{q_0\} \cup \{q_2\}$

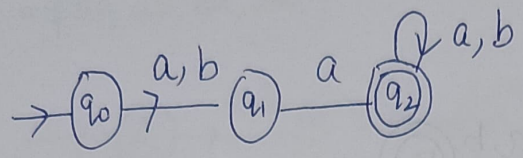
$= \{q_0 q_2\}$

⇓ Draw the dfa

Conversion of NFA to DFA

Ex: (4) An NFA accepting set of all strings whose second symbol from LHS is "a" over $\Sigma = \{a, b\}$

Ans



NFA

	a	b
→ q_0	q_1	q_1
q_1	q_2	Φ
* q_2	q_2	q_2

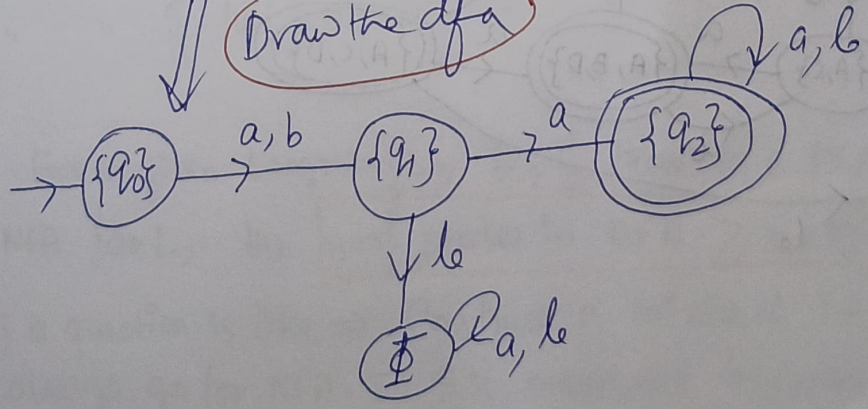
State transition of NFA

Use subset construction method

	a	b
→ $\{q_0\}$	$\{q_1\}$	$\{q_1\}$
$\{q_1\}$	$\{q_2\}$	Φ
Φ	Φ	Φ
* q_2	$\{q_2\}$	$\{q_2\}$

State transition of DFA

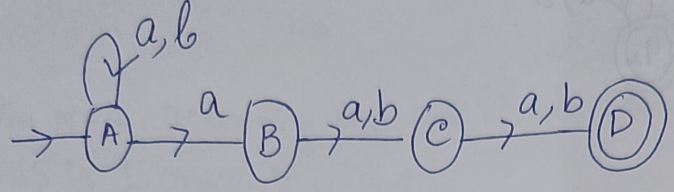
Draw the dfa



Conversion of NFA to DFA

Ex: (5) Construct an NFA that accepts set of all strings in which 3rd symbol from RHS is "a" or 3rd last symbol is "a" over $\Sigma = \{a, b\}$

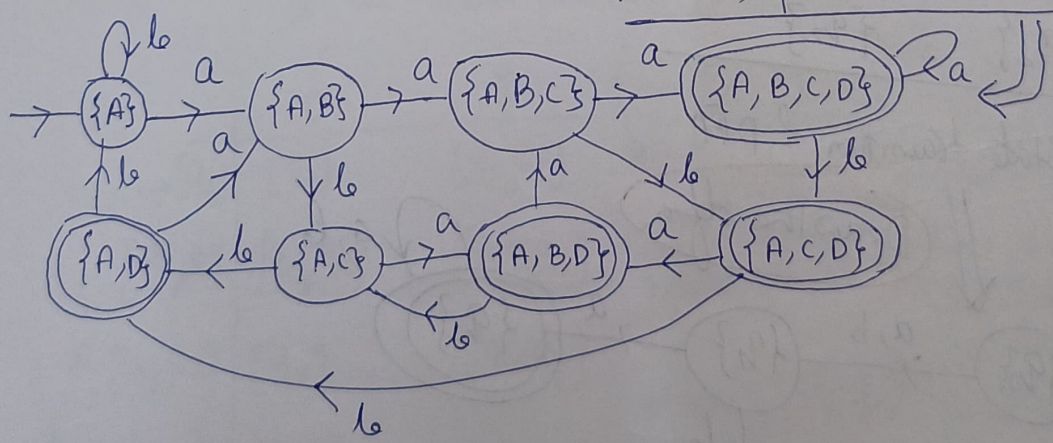
Ans



NFA's state transition diagram

	a	b
→ A	A, B	A
B	C	C
C	D	D
* D	Φ	Φ

Use subset construction ⟹

	a	b
→ {A}	{A,B}	{A}
{A,B}	{A,B,C}	{A,C}
{A,C}	{A,B,D}	{A,D}
*{A,D}	{A,B}	{A}
{A,B,C}	{A,B,C,D}	{A,C,D}
*{A,B,D}	{A,B,C}	{A,C}
*{A,C,D}	{A,B,D}	{A,D}
*{A,B,C,D}	{A,B,C,D}	{A,C,D}

Draw DFA